

Lab 3, Act 3

From counts to a continuous measurement

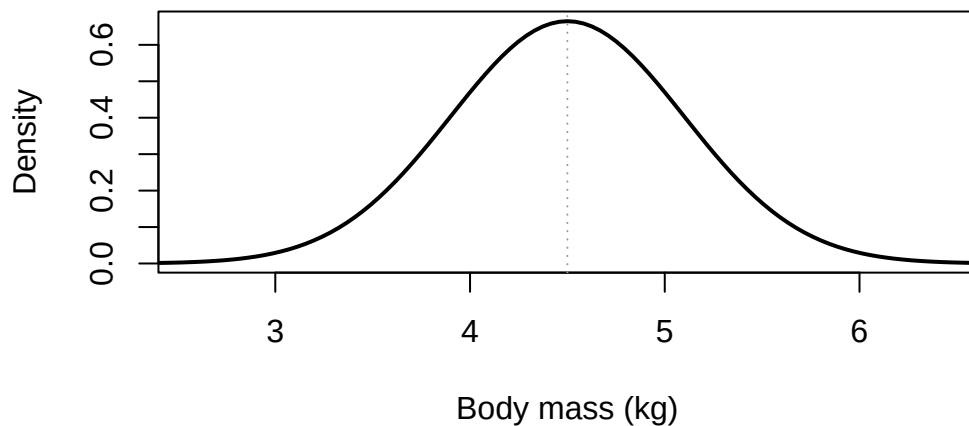
NRES 746 – name: _____ group: _____

Picking up where we left off

So far this lab has treated detections as discrete counts (0, 1, 2, ... detections). But some of the quantities the field team records are continuous. Suppose adult body mass for this carnivore, estimated from reference objects visible in camera-trap images, is well described by a normal distribution with mean $\mu = 4.5$ kg and standard deviation $\sigma = 0.6$ kg.

Part 1: Reading a density curve (individual, ~10 min)

Below is the probability density function (pdf) for this normal distribution.



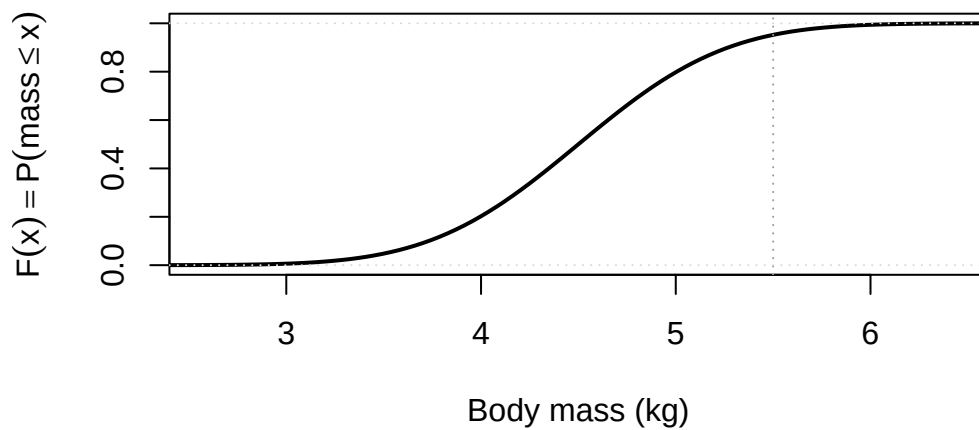
1a. Shade it by hand. On the curve above, shade the region that represents $P(4.5 < \text{mass} < 5.5)$.

1b. What is $P(\text{mass} = 4.5)$ exactly? Why – what's different about a single point on a continuous curve versus an interval?

1c. The height of the curve at mass = 4.5 is not zero (in fact, it's the highest point on the whole curve). If that height isn't a probability, what does it actually represent?

Part 2: Reading a CDF (individual, ~5-8 min)

Below is the cumulative distribution function (CDF), $F(x) = P(\text{mass} \leq x)$, for the same distribution.



2a. Using the CDF plot, estimate $P(\text{mass} > 5.5)$ (read off $F(5.5)$ as best you can, then subtract). Record your estimate – you don't need to be precise, just in the right neighborhood.

Group check-in (group, ~5 min)

Compare your shaded region (1a) and your estimate (2a) with your group and agree on a shared answer for 2a before opening laptops.

Laptops open now.

Part 3: Check your estimate (laptops, ~10 min)

3a. Use `pnorm()` to compute $P(\text{mass} > 5.5)$ exactly. How close was your group's graph-reading estimate from 2a?

3b. Use `qnorm()` to find the body mass value that marks the heaviest 10% of the population (i.e., only 10% of individuals are expected to be heavier than this value).

Group discussion and revision (~5 min)

Compare your Part 3 answers with your group. One thing worth discussing: `dnorm()`, `pnorm()`, and `qnorm()` all describe the same distribution but answer different questions – can your group state, in one sentence each, what question each one answers?

Feel free to revise any of your answers above based on the discussion (it's fine to cross out and rewrite – I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off going forward.
